LAB NO.05

Definition, Types, and Identification of Hard Disk Parts

1. Objective

To understand the definition of a hard disk, learn about different types of storage drives, and identify the main parts of a hard disk drive.

2. Apparatus Required

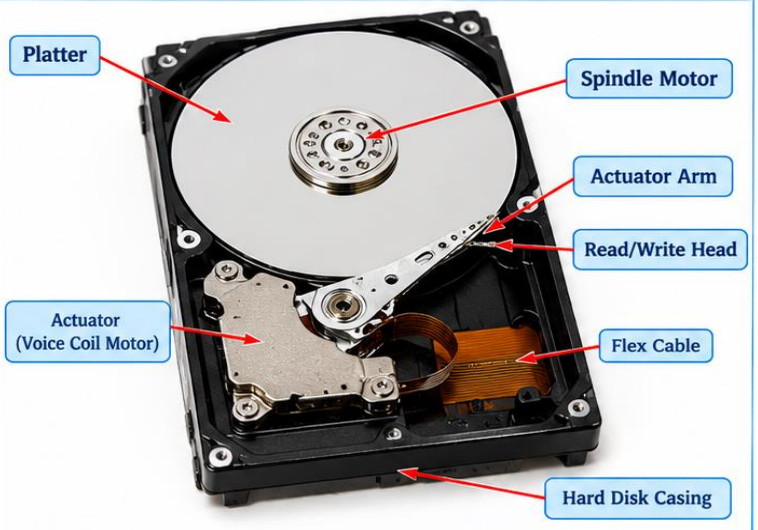
- Computer system
- Hard disk drive (HDD)
- Solid-state drive (SSD)
- Pictures or diagrams of hard disk parts

3. Definition of Hard Disk

A **hard disk** is a storage device used to store data, programs, operating systems, documents, pictures, videos, and other files. A traditional hard disk drive (HDD) stores data magnetically on rotating platters and retains data when the computer is switched off.

Identification of Hard Disk Parts
(Internal and External Parts)

1. Internal Parts of a Hard Disk (Open View)



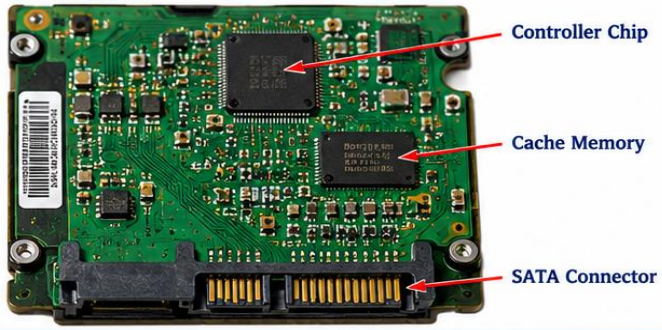
Parts and Their Functions

No.	Part Name	Function
1	Platter	Stores data magnetically.
2	Spindle Motor	Rotates the platters at high speed (usually 5400–7200 RPM).
3	Read/Write Head	Reads and writes data on the platter surface.
4	Actuator Arm	Moves the read/write head to the correct position.
5	Actuator (Voice Coil Motor)	Provides movement to the actuator arm using electric current.
6	Flex Cable	Connects the read/write head to the circuit board.
7	Hard Disk Casing	Protects the internal parts from dust, shock and damage.
8	Printed Circuit Board (PCB)	Contains the controller chip and connects to the computer.

2. External View of Hard Disk



3. Hard Disk Circuit Board (PCB)



Types of Hard Disk and Storage Drives

There are different types of storage drives used in computers. The most common are HDD, SSD, SSHD, and external drives.



1. Hard Disk Drive (HDD)

An HDD stores data magnetically on rotating platters. It offers large storage capacity at a relatively low cost.

Example: 500 GB, 1 TB, or 2 TB HDD.



2. Solid State Drive (SSD)

An SSD stores data in flash memory and has no moving parts. It is generally faster and more resistant to physical shock than an HDD.

Example: 256 GB or 512 GB SATA SSD.



3. Solid State Hybrid Drive (SSHD)

An SSHD combines a traditional hard drive with a small amount of flash memory to help speed up frequently used data.



4. External Hard Drive

An external drive connects to a computer, usually through USB. It is commonly used for backups, transferring files, and extra storage.



5. M.2 NVMe SSD

This compact solid-state drive connects through an M.2 slot and uses the NVMe protocol over PCIe. It can provide very high data-transfer speeds.

Comparison of Storage Drives

Feature	HDD	SSD
Technology	Magnetic platters	Flash memory
Speed	Slower	Faster
Moving parts	Yes	No
Noise	May produce noise	Silent
Cost per GB	Usually, lower	Usually, higher
Durability	More sensitive to shocks	More shock-resistant

Procedure

1. Observe the external appearance of a hard disk.
2. Identify the top cover and casing.
3. Locate the SATA data and power connectors.
4. Study a labeled diagram of the internal components.
5. Identify the platter, spindle motor, read/write head, and actuator arm.
6. Observe the circuit board and learn about its function.
7. Compare HDDs and SSDs.

Result

We learned the definition of a hard disk, identified different types of storage drives, and studied the main parts of a hard disk and their functions.